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Detecting terrorist influencers using reciprocal human-machine learning: The case of militant Jihadist Da'wa on the Darknet

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Over the past decade, social media has significantly impacted terrorism and counter-terrorism, serving as a platform for incitement to violence under the guise of religious preaching. This study explores the critical role of preachers preaching righteous behavior, a process known as Da'wa in Islam. Focusing on Militant Jihadist Da'wa calling to violence, the research analyzes 6000 posts from Darknet forums associated with Jihadist groups from 2017 to 2018. The study improves the detection and understanding of militant Jihadist preaching by using an advanced method called Reciprocal Human-Machine Learning. The study demonstrates the feasibility of better detection and a deeper understanding of influencing terrorists.

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Introduction

In the past two decades, social media (SM) has become pervasive in everyday life and, unfortunately, in the workings of terrorist groups too. Many terrorist groups have turned to SM to promote their agendas and, in particular, use influencers to disseminate the groups' propaganda with radical content (Stubbs-Richardson et al., 2020). They also use SM to recruit individuals to the group, to tempt people to donate, and to directly incite and glorify violent action (Aly, 2016). SM influencers, individuals who have significant followings in their networks, are usually effective in sharing text, pictures, audio, and videos, particularly with the younger generations (Woolley, 2022). Terrorist groups often utilize religious authorities as influencers in communities in which faith plays an important role. Our research examines how militant Jihadists use influencers on SM.

The move to the cyber arena by terrorist groups has significant implications on the dynamics of communication and influence in the group's networks. Firstly, as relatively small and nimble organizations, the terrorists recognize the disproportionate advantage that the technology affords them over their government and multinational rivals. Many terrorist groups are, therefore, committed to acquiring and leveraging the right technological and communication skills, often moving rapidly from one technology to another. Secondly, today's Internet opens communication to more participants and target audiences, to more styles of communication, to more media capabilities and formats, including audio, image, and video, and to more possibilities for crafting and tailoring the message. Thirdly, the new possibilities of adapting the media and the message, e.g., personalizing the message, afford the SM influencer a greater variety of messages and greater control over who gets which message. Personalized messages, for example, are often more persuasive when recruiting people. A fourth implication of the technology is greater flexibility in encrypting and securing sensitive messages. Furthermore, the move to different communication platforms, such as the Darknet, enables terrorist groups to evade or at least impede monitoring by counterterrorism organizations. Finally, the combination of new AI developments and communication technologies multiplies the effects of the above-mentioned dynamics. For example, message adaptation can be enhanced significantly with AI-augmented text classification of big-data messages and adapting the responses accordingly. Several terrorist groups have already used AI in drones, for example, and the potential misuse of AI by terrorists is threatening (Weimann and Weimann-Saks, 2024). Overall, the terrorists' use of the Internet and advanced technologies creates a new dynamic context that will require counterterrorism to adapt and overcome the terrorists' moves. Gaining a better understanding of the new methods terrorists communicate is therefore central to counterterrorism efforts (Ebner et al., 2022), yet elusive because of the continual changes in the communication patterns and ways of evading surveillance of SM by international security agencies (Koschade, 2006).

The new dynamics in the geo-political context in which terrorist groups operate create a need to continually learn the changing context and adapt counterterrorism activities accordingly. This need applies especially when using AI tools to combat the AI-augmented use of SM by terrorists. Indeed, as we argue below, these new dynamics reflected on SM dictate a dual research effort that combines two parallel and complementary directions: one is developing a deep understanding of how terrorists employ SM influencers to indoctrinate and incite terror, and the other is developing an AI-augmented model of learning from SM to serve as a general means of counterterrorism. Our research shows how the knowledge of SM influencers feeds into the AI-augmented model and how the knowledge grows based on feedback from the AI-augmented model. To gain a deeper understanding of the way this terrorist group uses SM influencers, we go back to the roots of Da'wa, which originally was about calling and convincing others to accept the Islamic faith. In section B1 below, we trace the evolution of Da'wa's target audience and use of media till its practice today by terrorists, concentrating on militant Jihadists using the Internet (Lewis, 1993).

For the second research direction in this paper, we develop and tailor a new research method to fit the technological and social context of militant Jihadists on social media. It relies on big data and AI techniques needed to analyze communication on SM but also on human experts with a deep understanding of the social dynamics of the terrorists' interactions in each setting. Approaches that have combined linguistic processing with behavioral cues have shown some success (Thorley and Saltman, 2023). Although there is promising research (Bérubé et al., 2023) that uses socio-semantic network analysis to help identify individuals who incite violence in radical networks, deriving a social network for analysis is not always feasible. In assessing anonymous and disparate communications forums, as we will discuss, there are no social network structures to analyze and must rely solely on linguistic analysis. Therefore, we – the research group, pursue a context in which most actors of the social network are unidentifiable, such as actors communicating on the Darknet.

Section B2 lays out the challenges of automated detection by examining the discourse that exists on Darknet platforms, which must consider the *context* of the Da'wa institution and practice (developed in section B1). To face these challenges, we use a new reciprocal human-machine learning (RHML) methodology in which the human expert and the AI model learn reciprocally from each other to detect Jihadist Da'wa in a dynamic discourse on SM. This approach enables both human experts and automated machine-learning models to improve their performance over time (Schwartz et al., 2023).

In method section C, we explain how the RHML methodology can be applied to detect terror-related communications. In the initial stages of our research, we found that in order to study terror-related communications as it is manifested in the discourse on SM, we must distinguish between messages that pertain directly to terrorists or to the commission of acts of terrorism, namely Jihadist Terror Activists (JTA) and messages that pertain to preachers who call upon others to commit violent acts in the name of Islam, namely Militant Jihadist Da'wa (MJD). Consequently, we developed a classification method that differentiated between JTA and MJD, where the latter often precedes the former.

The results section D describes how we applied the RHML methodology to two forums that are used by terrorist groups and demonstrates the stages in which the expert and the machine learn reciprocally to detect MJD from each other, step by step. In sections E and F, we present a discussion of lessons learned and limitations and a conclusion with recommendations.

Theoretical framework for studying Da'wa, Militant Jihadi Da'wa and automated identification

B1. From Classic to Militant Jihadist Da'wa on the Darknet. A core characteristic of religious terrorism is the communication of sacred context by clerical authorities (Bonner, 2006; Ahmad and Azzam, 1976). Hence, the critically important role of clerical sanction in legitimizing and motivating religious believers to perform acts of terrorism. Moreover, the particular nuances of communicating the religious messages, as well as the media used, require a deep understanding of the particular stream of religion in question. Therefore, this section highlights the important role

of domain experts in characterizing the differences between various groups and defining types of Da'wa. These insights accompany the entire study and emphasize that good automation results from collaborative work between a quality domain expert and advanced computing capabilities.

The word "Da'wa" (دعوة) derives from the Arabic verb Da'a (داع) which means to announce, call, and invite; Da'wa activity is defined as calling and convincing others to accept the Islamic faith. A person who performs Da'wa is called a "Da'i, داعي" (singular) – "Du'āh, دُعَاة" (plural) meaning an Islamic preacher. The people defined as Du'āh have deepened their understanding of Islamic teachings and are committed to sharing their knowledge with others (Ahmad and Azzam, 1976). Moreover, it may stem from the Quran that a Da'i must also accept three elements to be bestowed with the title: knowledge, etiquette, and politeness in dialogue (Quran: 16:125). A Da'i, therefore, can be any Muslim who uses arguments based on accepted Islamic sources to preach to others to embrace Islam and join the 'Umma' (اَلْأُمَّة), the community of believers.

Da'wa has existed since the beginning of Islam as primarily a non-violent call to non-Muslims to embrace the religion of Islam. Some early exceptions are notable, such as that of the Ismailis (Daftary, 2011) and the Abbasids during the rule of the Umayyad House, whose purpose was to bring about a coup against the existing ruler (Lewis, 1970). Over the years, starting with the Ismailis, who were the first to do so, Da'wa also aimed at Muslims and invited them to keep to the straight path as well as to persuade Muslims to follow the lead of Islamic groups or different movements.

In the 21st century, Da'wa relies not only on classical face-to-face preaching but also on digital preaching on various platforms. One of the leading platforms for disseminating Da'wa by the new Islamic preachers are social media platforms such as Facebook, Twitter, YouTube, and other internet forums (Malik and Weismann, 2020). Nevertheless, these channels are not safe for MJD because anonymous individuals seeking to carry out Islamic terrorist activities or join Islamic movements are regarded as illegal by various international authorities. As a result, the need has arisen for a different platform – such as the Darknet – that can hide preachers from the authorities or intelligence services (Weimann, 2019).

To hide the identity of the group's supporters and the content disseminated after the 2015 Paris terror attacks, ISIS turned to the Darknet and other hidden social networks such as Telegram. This move made it difficult to monitor terror-related communications on social media. Three years later, in April 2018, a report called "Terror in the Dark" exposed the growing use of the Darknet by terrorist groups, primarily for propaganda and active recruitment purposes. The findings demonstrated how terrorists and extremists created a large number of safe havens in which they hatched plots, raised new donations, and recruited followers (Malik, 2018).

MJD of the past 30 years, the subject of this study, is located, to a large extent, on the Internet. This particular variant of Da'wa deals with messages that are usually disguised and encrypted, complicates the analysis even further and often triggers countermeasures from the messengers, such as moving from open social media to hidden platforms such as the Darknet.

Table 1 summarizes how Da'wa was disseminated from Islam's earliest days through to MJD in the 21st century (in the last row of Table 1). Table 1 shows the evolution in three dimensions: the media used, the messenger, and the audience. The table presents the transition from face-to-face mass communication to print and digital media. The new media challenges the concept of direct contact between the preacher and the addressee while creating more opportunities for direct discourse with the new social interactions created by technological changes (Ridho, 2019). These changes culminate in the characterization of MJD in the 21st century as it functions on social media.

We define MJD as the dissemination of messages that call for militant Jihad. The term Jihad has various meanings, and Muslims themselves can regard every daily activity as Jihad (Streusand, 1997). MJD is a form of recruiting used by terrorist groups, mainly of the Salafi-Jihadist stream, to change reality through violent means (Maher, 2016). Salafia-Jihadiya is a modern term and has become prevalent since the 1990s (Lauzière, 2015). However, the Salafia-Jihadiya worldview has its roots in the neo-Hanbali stream. For example, the important thinkers Ibn Taymiyyah and even more so Ibn Abd al-Wahhab abided by this view (Cook, 1992). In this context, the division of Salafists is

Table 1 The Evolution of Da'wa Transmission, From Its Genesis to Modern Times.

Content	Media	Messenger	Target Audience	Era
External call to join Islam	Face-to-face	The Prophet Muhammad and the "Rightly Guided Caliphs" (Culapha al-Rashidun).[a]	Arab peninsula residents and idolaters	7th–9th centuries
Internal call for correction of society (mostly Shi'ite)	Face-to-face	Shi'ite Sages.[b]	Shi'ites and Sunnis in the Middle East	9th–12th centuries
Internal and external call	Face-to-face	Sunni Sages, particularly from the Salafi and Wahhabi.[c]	Muslims, including Shi'ite and Sufi	13th–18th centuries
Internal and external call	Face-to-face + print	Muslim scholars (khatibs, imams).[d]	Muslims and non-Muslims	18th–20th centuries
Internal call	New forms of media, TV, digital and print	Muslim scholars (khatibs, imams).[e]	Muslims	20th century
Internal call	Social networks	Influencers, any Muslim.	Muslims and non-Muslims (global)	21st century
Internal and external call	Hidden networks such as the Darknet and Telegram	Influencers, any Muslim, anonymous Muslims.	Muslims and non-Muslims (global)	21st century and currently

[a]For further information, refer to: Ghetas, Hosni Muhammad, Al-Da'wa Al-Islamiyah fi 'aed Amir al-muaminin Omar Ibn al-h'atab rad'a Allah 'ano. Cairo:Al-zarah lil-i'lam al-Arabi. 1995. p 90.
[b]For further information, refer to: Daftary (2011) The Isma'ilis: Their History and Doctrines. Cambridge University Press. p 304.
[c]For further information, refer to: Wiktorowicz, 2006 The management of Islamic activism: Salafis, the Muslim Brotherhood, and state power in Jordan. SUNY Press.
[d]For further information, refer to: Amin sazik, Al-Da'wa Al-Islamiyah: farida shariya wa-d'rura bashariya, *Oman. Jamihat a'mal al-mtabia'* al-t'awtiya. 1982. P.15. And: Mohammed al-Ghazali, Al-Da'wa Al-Islamiyah tastakbilu kurnaha al-h'amis 'ashar, Damascu: Al-Ghazali, 2001. 2nd. Ed. p 45.
[e]For further information, refer to: Jakob Skovgaard- Petersen, The Dai in Contemporary Arab Television Fiction: From "The Foremost of Preachers" to "Our Master". In: Malik & Weismann (eds) (2020). Culture of Da'wa: Islamic preaching in the modern world. The University of Utah Press. p. 209.

mainly into three streams: Purists, Politicians, and Jihadists (Wiktorowicz, 2006). The Salafists share the same doctrine (aqīdah, عقيدة) and similarly consider innovation forbidden (bida, بدعة) and accept the laws of Shari'a and precepts of Da'wa as guiding principles in their daily lives. However, each stream applies its own teachings (manhaj, منهج) in different ways.[1] The Da'wa of the third group – the Jihadists – promotes 'Jihad takfiri' against the "infidels". Additionally, the Jihadist preacher is motivated by an ideology that states that attacks on the West or apostate Muslims are necessary to achieve worldwide Shari'a dominance (Hoffman, 2020).

MJD can convince and demonstrate credibility by relying on references from religious sources to justify their arguments (Weismann, 2020). Two prominent examples are (1) Interpretation of the verse "Commanding Right and Forbidding Wrong"[2]. (2) Rendering Da'wa and Jihad to a personal duty (fardʿayn) rather than a collective duty (fard kifaya). For example, in terms of Jihad, the writings of the theological founders of the al-Qaida organization[3], as well as Muhammad Abd-al-Salam Faraj's book "The Neglected Duty" – "al-Farida al-gha'iba" (Jansen, 1986). An example in terms of Da'wa is the "Muslim Brotherhood" theological founders' writings.[4]

Afterward, the same ideas were conducted online. A prominent example of a Jihadi preacher who was previously a senior agent in Al Qaeda is Anwar al-Awlaki. Al-Awlaki argued that the role of a preacher was to spur potential activism and make activists believe that militant Jihad is a personal duty (fardʿayn) rather than an act of terrorism (Alshech, 2014). Research published on al-Awlaki shows us that the website he established, 'Internet Mujahidin,' was used as a platform for Da'wa and gave Al-Qaeda an advantage in recruitment (Conway, 2012).

Fahad Qureshi is another example of a preacher who utilizes the Internet as a tool to call on believers to perform Da'wa. Qureshi published a video entitled 'Da'wa is Our Identity' on the *IslamNet* YouTube channel. It became one of the organization's most highly viewed videos. In the follow-up YouTube video entitled 'Work with Da'wa', Qureshi claims that Da'wa is not just a personal duty-fardʿayn - but also a collective duty - fard kifaya (Bangstad and Linge, 2015).

B2. The Rise of the Darknet and Automated Detection. The Internet, which originated as a communications project of the US Defense Advanced Research Projects Agency (DARPA) in the 70 s, provides a communications transport layer supporting data transmission through what became the TCP/IP protocol. Since then, it has evolved to support the use of many different forms of communication. The part that is easily accessible to all, such as unprotected websites serving content through the HTTP protocol[5] (Berners-Lee et al., 2010). The Deep Web generally refers to password-protected databases and information resources that remain connected to the surface web through the HTTP protocol but can only be accessed by using the correct predetermined username/password credentials (Rosenzweig, 1998). This mainly consists of private network websites and secure sources inaccessible to the general public, housing private data such as bank accounts and medical information (Weis, 2010). The Darknet refers to other messages on the Internet that are accessible only through more obscure protocols such as Tor (The Onion Router) (Schwartz, 2010).

Terror organizations initially shifted from the surface web to the Deep Web to benefit from password protection and eventually to the Darknet due to increased law enforcement activities (Gendron, 2017). As automated detection processes were developed and applied to Darknet data, the combination of two factors led to the rise in the development of new methods for automated detection, much of which has focused on analyzing data traffic patterns (Fachkha and Debbabi, 2016). The first factor began in the early 2000s, following the terrorist attacks of September 11, when there was a significant increase in research deliberating the capacity to identify terrorist threats in an automated manner (Ganor, 2019; Medina, 2014). The second factor began following the rise in the status of social media, particularly on platforms such as Twitter (Berger and Morgan., 2015; Zelin, 2014) and Facebook. In its wake, another wave of literature emerged that sought to offer methods to identify terrorism (Hashemi and Hall, 2019; Wolfowicz et al., 2021), and Jihadism (Weimann, 2006; Weimann, 2010). Alongside the literature covering attempts to identify moderate and extreme sentiments of the Jihad phenomena (Silverman and Sommer, 2019), many studies also discussed techniques to identify extremism on social media (Saif et al., 2017; Ferrara et al., 2016).

B2.1 Automated classification and classification measures. While most automated classification tools used to detect suspect messages on SM are based on AI text classification models (Wignell et al., 2021) we argue that continuous human insight based on a deep understanding of MJD and JTA generated by human experts is needed to complement and adapt the AI models. In other words, we advocate AI-augmented classification rather than AI-automated classification. In building our classification system, we make extensive use of machine learning models – in particular, Bidirectional Encoder Representations from Transformers (BERT) - which we assess using measures such as Area under the Curve (AUC). Both of these terms are now introduced briefly, with pointers to a more extensive background.

BERT models: The Bidirectional Encoder Representations from Transformers- or BERT is a powerful natural language processing (NLP) model that uses a deep neural network architecture to represent language. Developed by Google, BERT algorithms are based on the state-of-the-art transformer model. When properly trained, it becomes the basis for automated language understanding (Devlin et al., 2018).

Performance measures: The Area under the Curve (AUC) is a metric used to evaluate the performance of a binary classification model, such as BERT, in a classifier context. The Receiver Operating Characteristic (ROC) curve is a graphical representation of a classifier's performance, with the AUC representing the area under this curve. The ROC curve plots the true positive rate against the false positive rate at various classification thresholds (Huang and Ling, 2005).

For a BERT classifier, the AUC is computed by varying the threshold for classifying instances as positive or negative and calculating the corresponding true positive rate and false positive rate. A higher AUC value indicates better discrimination between positive and negative instances, reflecting the model's effectiveness. Essentially, AUC provides a summary of a classifier's ability to distinguish between classes across different threshold settings, offering a comprehensive evaluation of the model's overall performance in terms of sensitivity and specificity (Shishah (2021)).

B2.2 An AI-augmented model - Reciprocal Human-Machine Learning. We build on an AI-augmented text classification model that incorporates human expertise in the learning loop to promote continual learning for both humans and machines, which is described in detail by Te'eni et al. (2023). By combining machine-learning ML algorithms for text classification and human sense-making processes, the RHML model is used to detect MJD suspects from text messages on Darknet forums.

Figure 1 depicts the essence of the RHML model, which is a learning loop in which the human expert trains the machine-learning algorithms but at the same time also learns from the machine how to classify messages that indicate a suspect. Such a learning loop is needed, especially when the content and form of messages through which MJD and JTA are conducted and communicated change continuously and when the classification knowledge must be updated accordingly.

To measure the dynamics of the learning loop, we use AUC as a measure of classification performance (accuracy) and the complexity of decision trees as an indicator of the expert's classification knowledge.

Research Methodology

Over the two-year course of this research project, it has progressed through the following five stages: a) collecting data from the Darknet; b) delineating the classification question; c) initial text classification; d) establishing an initial conceptualization of MJD and JTA; and e) performing the iterations in the RHML loop to refine the initial model for MJD and JTA.

I. **The first stage** was to find suitable forums from which data could be extracted. The research group partnered with the Cyber Threat Intelligence Company Cybersixgill[6], which mined the data to ensure the content remained in the form of original data. The researchers provided keywords that characterize Salafiya-Jihadiya ideology (detailed in the following endnote).[7] The list of keywords brought up several Darknet forums from which the domain expert selected two for analysis: "*Forum_-leyoth*" and "*al-Fidaa*". The research content focuses on the period between 2017 –– and 2018, which marks the ISIS wars in the Middle East region and the onset of the organization's weakening.

II. **In the second stage**, the research group defined the classification question (Osisanwo et al. 2017). To this end, 3,000 posts were sampled from these two chosen forums. Then, the domain expert divided the forum users into three subsets: Jihadist Terror Activist (JTA), Militant Jihadist Da'wa (MJD), and Non-suspect (NS). In the posts of all three subsets, the users glorify ISIS conquests and escalate or exacerbate the discourse against 'infidels' or 'sinners' by using relevant Islamic terms (such as murtidun - مرتدون, shirq شِرك -). However, the initial differences identified between them were as follows: JTA promotes terrorism and is directly involved in the conflicts waged by ISIS, whereas MJD is distinguished by writing more in-depth texts, a distinction which is reflected in the quality of their writing. The domain expert established a classification table as part of the expert's conceptual model (see Table 2).

III. **In the third stage**, an additional 3,000 posts were tagged, for which the domain expert marked texts identified as 'JTA suspect', 'MJD suspect', or 'NS' (Non-suspect). The research group re-examined whether the insights produced in the first stage were still valid. In addition to validating the insights, a decision tree was designed to capture the experts' insights from the texts. At the beginning of this stage, the initial decision tree was relatively simple, and it grew with each iteration to include more diverse decision rules. The initial decision tree is shown in Fig. 2. The difference between the initial and final decision trees shown in Fig. 3 demonstrates how the researcher's learning developed as the research progressed.

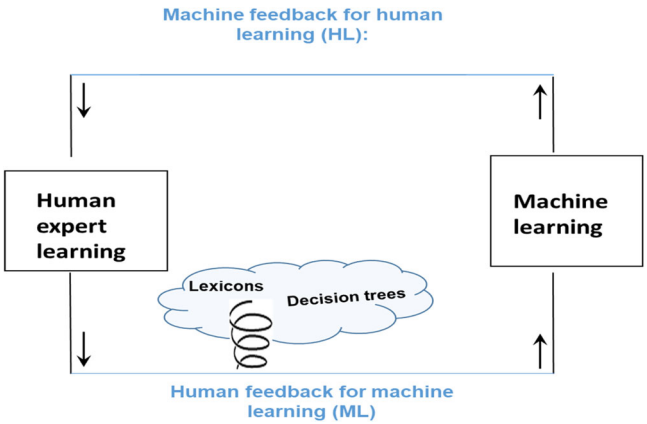


Fig. 1 The learning loop in Reciprocal Human Machine Learning (adapted from Te'eni et al., 2023).

In the fourth stage, we defined the unique features of content attributed to JTA and MJD that can increase machine classification accuracy. The data scientists implemented a baseline deep-learning (automated) classification model based on Arabic-BERT (Safaya et al., 2020). Arabic-BERT is a language model with BERT architecture pre-trained on a large corpus of Arabic for fill-in-the-blank tasks. The data scientist fine-tuned two different general-purpose Arabic BERTs: one with the **Qur'an** and the **Hadith** corpus (Literature BERT below) and another with a **news corpus**

Table 2 Formulation of Militant Jihadist Da'wa versus Violent Jihadist Terror Activist.

Characterization of Suspects

Militant Jihadist Da'wa (MJD)	Jihadist Terror Activist (JTA)
Understanding and expertise of the Hadith literature	Talking about acts in the name of Islam and uprising.
Understanding and expertise of the Holy Qur'an	Praising acts of terror (mainly in ISIS texts).
Preaching for daily Islamic commandment (i.e., modesty, five pillars of Islam), using nuances from Islamic law. And using Islamic months in the writings, i.e., the Hijri calendar.	Denunciation of unfaithful Muslims using proper Islamic condemnations (مرتدون, شِرك)
High-register literary Arabic (called fuṣḥá [الفصحى]) or scholarly language.[a]	Understanding of geographic zones and locations of the battlefields.[b]
Salafist Jihadist terms used by both users: آل سلول, لتكون كلمة الله هي العليا, الشيخ العلوان حفظه الله, تمييع الدين, صنم الوطنية, الديمقراطية صنم العصر.	

[a]High-register literary Arabic (fuṣḥá) manifested in the suspect writing style is defined as follows: using words in literary Arabic rather than spoken Arabic or correct spelling. For example, "حلو", and not "حلوووو", correct vocalization of verbs, abiding by syntax rules, using conditionals, future rules (إن, كان), verbs must be written before their equivalent noun, etc.
[b]Knowledge of geographical zones mentioned in the suspect text can be used in the following way: a report from the battlefield in Mosul (الموصل) means that the user is in Iraq; likewise, a report from Raqqa (الرقة) means that the user is in Syria, etc. These texts describe a contemporary occurrence of combat action in a particular place, as opposed to news reports that describe a past situation.

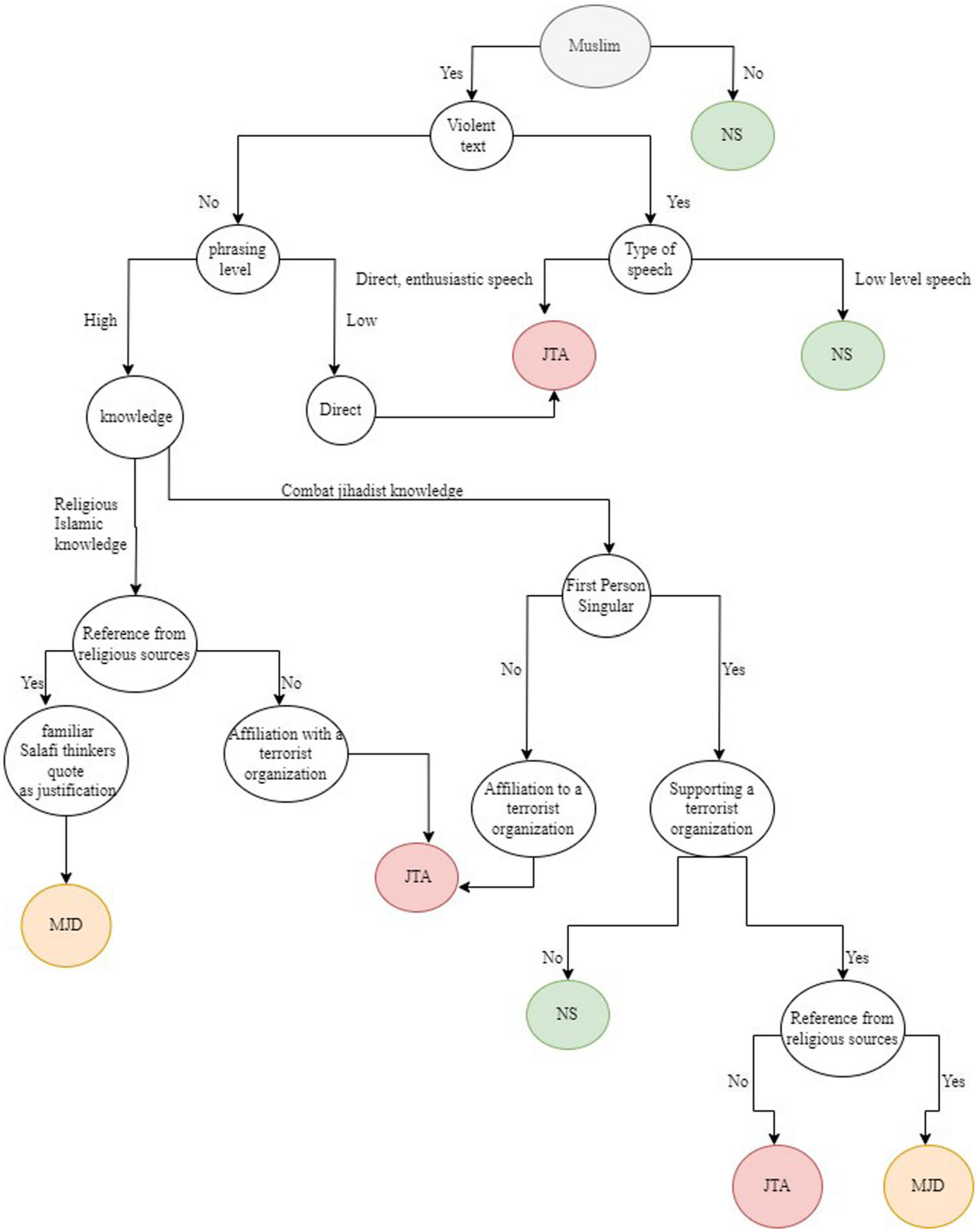


Fig. 2 Initial Decision Tree of Expert's Classification Model.

on the topic of ISIS taken from Al-Arabiya, a news channel based in Dubai [8] (see Table 3).

IV. **In phase five**, we documented the lessons learned from the RHML-based research process. The quantitative and qualitative questioning of *why* the machine algorithm was wrong – and the attempts to improve it – ultimately became successful thanks to the learning loop discussed in the following section –

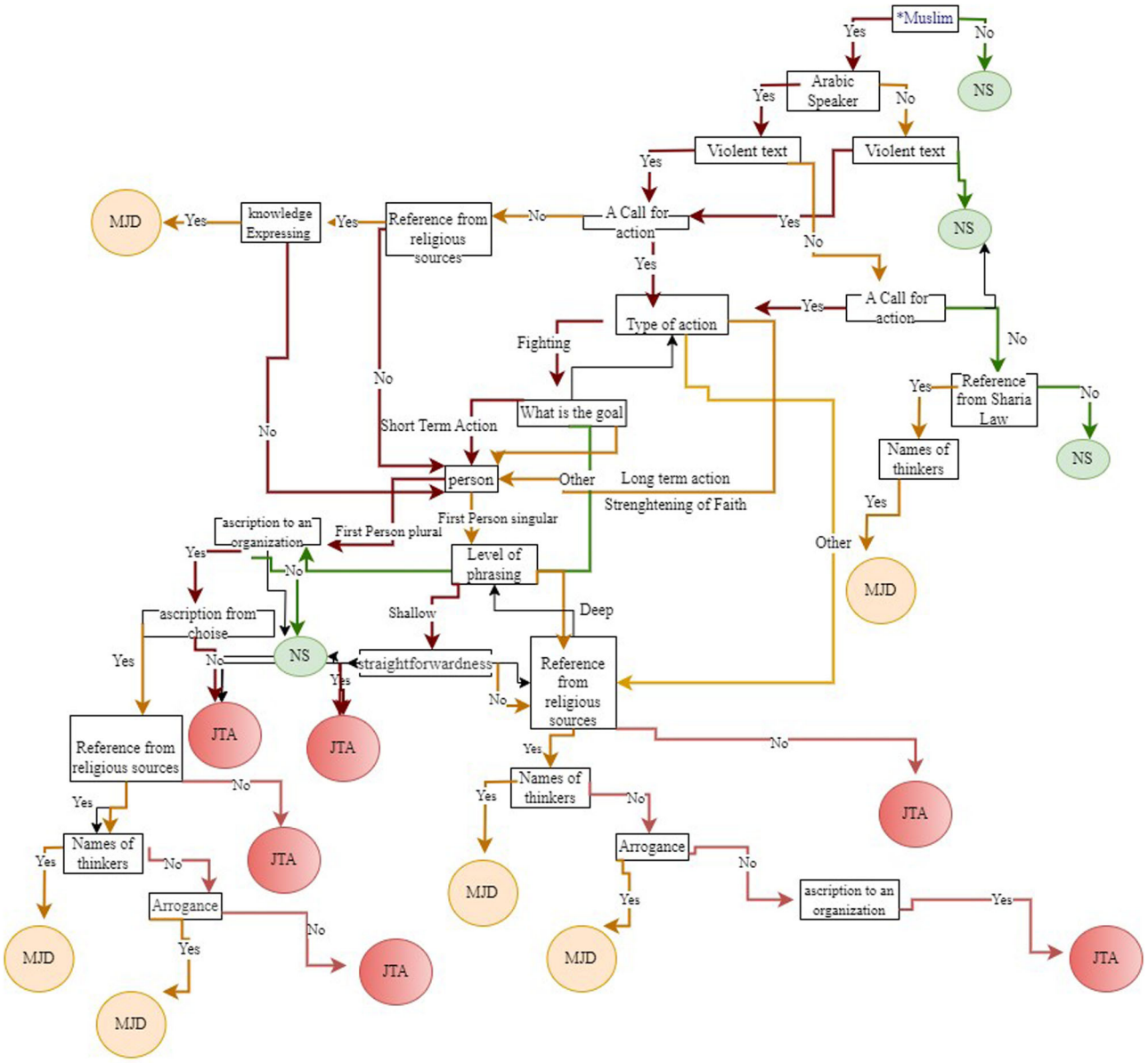


Fig. 3 Final Decision Tree of Expert's Classification Model.

Table 3 BERT Performance After Human Insights that Can Improve Precision of Machine Learning.

Classification Accuracy for Non-suspect versus Suspect (JTA or MJD)		
	JTA	**MJD**
General purpose BERT	0.71	0.65
Literature (Qur'an, Hadith) BERT	0.69	0.70
News (Al-Arabiya) BERT	0.73	0.69

Results

The results of applying the RHML to detect Jihadist Da'wa can be seen as two parallel learning processes, namely, human and machine learning, to classify. Although the feedback from one process to the other affects the results of the other, we present the progress of each separately and later discuss the reciprocity.

The steps of the continual learning loop can be summarized as follows: (1) Human expert generates a conceptual model (in the form of a decision tree and lexicon); (2) Machine accepts feedback from the expert (elements of the conceptual model); (3) Machine runs classification algorithms based on tagged posts; (4) Machine returns feedback to the human domain expert; (5) human domain expert reads the feedback and manually revises the conceptual model. There were five interactions between the domain expert and the "learning machine", detailed as follows:

1. In the first stage, the domain expert distinguished between two types of terror-related suspects – JTA and MJD (see Table 2). The data scientists trained the classification model using BERT model in Arabic (the ML precision was 65%).[9]
2. In the second stage, the domain expert observed a linguistic difference between literary Arabic texts and spoken Arabic texts. JTA mostly used spoken Arabic, which can involve mistaken grammar and double-letter writing. On the other

hand, MJD used high-register literary Arabic, which demands attention to Arabic syntax and grammar rules as well as knowledge of the Qur'an language. An example of a grammatical rule is to put the verb before the subject.[10] This led the data scientist to include a feature in the model referred to as "parts of speech" aimed at identifying their role in the sentence (Abdelali et al., 2016).

3. In the third stage, we observed yet another difference in the content of the posts. In many cases, the MJD quoted references from the Shari'a sources (Quran and Hadith literature) to enhance and base his statements.[11] To increase the precision, the model was trained on a database which consisted of Shari'a sources mainly used by scholars that are commonly cited by the Salafi Jihadi movement[12]. In this context, it is important to emphasize that Jihadist thinkers mainly rely on verses that date back to the Al-Medina era and not the Mecca era since, at this point, Islam was at the top of its conquests (Mozaffari, 2007). Due to the model training, the precision rate (AUC) among MJD increased to 71%. (see Table 3)
4. In the fourth stage, the domain expert noticed that JTA tends to post geographical locations of active battle scenes. At first, the domain expert gave the data scientist a list of common terms mostly used by terror suspects in forums[13] as well as an ISIS online battle map, which was relevant to the time of publications surveyed in the forum.[14] However, the data scientist examined the use of automatic extraction of Arab locations via a pre-trained language model (Benajiba et al., 2007). He then used these locations as indicators of JTA. [15]
5. At the fifth stage, as we studied the labeling of the model, the research group noticed that the model characterizes news reporters as JTA. At first, the domain expert attempted to use a self-made lexicon with words characterized by news sites. Nevertheless, the data scientist decided to keep the conceptual way of machine learning.[16] As such, the domain expert suggested that the model trains on well-known news sites of ISIS vis-à-vis legitimate news sites, and so the model will recognize the differences.[17] Having integrated the above-mentioned features, we re-examined examples of 150 posts. We discovered that the domain expert misclassified eight posts. Demonstrating the utility of the machine detecting human error.

We note the following reservations in this process: (1) There is still knowledge that the research group did not manage to transfer to the machine and convert the insight to a feature in the model. For example, among MJD, the use of historical events as a reference and justification (The Battle of Badr, Treaty of Hudaybiyyah, Battle of Khaybar) or, for example, the use of the Hijri calendar, which indicates a stronger religious affiliation. (2) Following the use of Darknet, we cannot tell whether a certain user is really present in the battle zone and reports from the field or whether the report was sent from a different user and the former only posted. That is another reason why we focus on the content rather than the user.

Table 2 organizes the classification knowledge after the second stage of the methodology according to the characteristics of each type of influencer as defined by the expert and fed into the machine algorithms. This characterization is divided into binary features that differentiate these two groups from other participants active in these forums so that at the end of the process, the machine was able to identify them specifically. (For examples of message classification accordingly, see Tables A1 and A2 in the Appendix). Human learning to classify MJD or JTA is shown by the difference between Fig. 2 (initial decision tree) and Fig. 3 (final decision tree after stage 4). Quantitatively, Fig. 3 has almost three times more concepts than Fig. 2. Regarding the relationship between these concepts, in Fig. 3, there are almost four times more connections than in Fig. 2. (See calculations in Table A3 in the Appendix). The decision tree also explicitly makes the working assumptions for classification explicit. For instance, the forums are written in Arabic and have members who identify as Muslims. Therefore, the first condition was that the research subject should be someone who is undoubtedly identified as a Muslim and writes in Arabic texts that reflect violent aspirations.

Likewise, as seen in the decision tree, one dereference phrasing is listed as "shallow" or "deep." This means that similar calls for violent actions in the name of Allah or preaching do not carry the same level of influence but must be discerned based on writing style. For instance, the writing style of JTA reflects spoken Arabic rather than literary Arabic; there is no reliance on scholars, references, or evidence from Shari'a law that reflects religious knowledge. For JMD, the opposite is true, and a relatively high register of literary Arabic (fuṣḥá) is used compared to other users in the forums. The writing style of the MJD exhibits proficiency in Islamic literature (the Hadith and the Qur'an) and primarily preaches for Islam to exhort others to be modest and fulfill the five pillars of Islam. The JTA talks more about acts in the name of Islam and describes contemporary occurrences of combat action in a particular place.

The result of applying RHML is a higher accuracy (AUC) due to human insights. Table 3 shows the improved performance detecting suspects for Jihadist Terror Activist (JTA) and Militant Jihadist Da'wa (MJD), according to the corpus used by BERT.

Discussion

The widespread use of Internet triggered a general discussion about religion on digital media and the challenges associated with it (Berlinger and Te'eni, 1999). Islam was no exception. In the past, Muslims seeking to engage in Da'wa traditionally adhered to three fundamental elements from the Qur'an: knowledge, good manners, and courteous dialog. In comparison, today's online Da'wa lacks clear characteristics. Its transition to social media has been criticized for several reasons, such as loss of intimacy, inaccuracy, and even 'lack of authenticity' (Ermayani et al., 2021), as have other online services in other religions (Mubarak, 2022).

Nevertheless, online Da'wa continues to spread and has become a major component of Islamic political thought. In the context of this research, a Militant Jihadist Da'i is categorized as an individual with deep knowledge of Islamic tradition, law, and theology who is highly proficient in literary Arabic (Fuṣḥá [الفصحى]) and who quotes scholars that are commonly cited by the Salafi Jihadi movement. The Jihadist preachers thus gave legitimacy to online forum activists who sought to use Islamic sources as references justifying the use of violent activity against non-Muslims or Shiite and Sufi civilian Muslim populations.

As argued in this paper, MJD is a preliminary stage that precedes militant Jihad or terror activities. MJD is capable of indoctrinating believers into carrying out terrorist acts while simultaneously leading believers to think that they are protecting the nation ('umma) and performing a religious duty (fardi'yan). This study has demonstrated how a qualitative researcher can characterize militant Jihadi discourse, label it, and then provide the machine with inputs that will improve the quality of automated identification.

This research is motivated by three factors that emerge from the dynamic nature of terror communication patterns on the Internet. The first is the long tagging process in which a domain expert may lose attention. The second factor lies in the ongoing transformations that are wrought by changing times and technologies, which result in the transition to different platforms. The third factor is the vibrant language and culture dynamics, particularly in the streams of Islam that are exposed to new preachers and additional fatwas (Islamic laws). Due to the above-mentioned

factors, our research found that it is essential to provide the domain expert with automated tools to which insights may be added. The machine and the human expert will continue to learn and adapt in response to the unique circumstances of modern times and technology. This research suggests that future studies implement the same methodology adopted in our study while dealing with various social platforms.

Conclusions

This study makes two main contributions. The first contribution lies in the fact that, to date, the effort to identify terrorism suspects has focused predominantly on the operational side—that is, the terrorists themselves. In contrast, this article addresses those who influence and propagate militant Jihad. We distinguished between MJD (Militant Jihadist Da'wa) and JTA (Jihadist Terror Activist). This was followed by the development of an orderly methodology; a chart demonstrating the differences between the users was provided so that follow-up studies could add to and deepen the characterization. Decision trees and tables showing the process of the study and of the qualitative analysis were also included. As for the latter contribution, the machine began by learning how to conduct qualitative labeling. The need to raise the precision rates of machine identification was then established. Following a team discussion, this need was implemented by inputting Islamic texts into online corpora, including the Qur'an and Hadith literature, as well as Salafi thinkers' publications that have been popularly quoted by Jihadist preachers.

The second contribution is the ability to analyze the evolving discourse of Jihadist Da'wa on social networks, which can assist researchers in the field. As we demonstrate, this ability requires a new research methodology based on reciprocal human-machine learning (RHML), which is a collaboration between a domain expert and a machine-learning text classification method. The research team incorporated data from well-known news sites run by ISIS and al-Qaeda to allow the machine to be able to distinguish between news items and JTA reports from the field. In this way, the RHML learning loop ultimately experienced a double improvement in both the machine's results and the qualitative researcher's understanding.

Data availability

The dataset generated and/or analyzed during the current study is submitted as a supplementary file and can also be obtained from the corresponding author upon reasonable request. The files submitted as supplementary material contain approximately 1170 tagged posts from the "Al-Fidaa" forum (from 2018) and the "Leyoth" forum (from 2017). The data are presented in its raw form, with the main features described in detail within the body of the article and through examples in the appendix. In the Excel files, you will find the tagged posts, their content, and the time-stamps indicating when the posts were created. Regarding the data mining process, as detailed in the methodology section of this study, the research team first determined that the platform to be examined would be the Darknet. To this end, we contacted a third-party company named Sixgill. We then compiled a list of keywords (Endnote 12) designed to surface data relevant to our research, specifically focusing on militant Jihadist forums in Arabic. After reviewing several examples from different forums, the research team selected the forums discussed in this study because they met our research criteria. We received the data in Excel files as presented and subsequently began the tagging and analysis process. The remaining steps of the methodology are detailed in the corresponding section of the article.

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Notes

1 The Purists do not wish to assimilate into Western society, and so they carry out their Da'wa activities primarily among Muslims. In contrast, the Politicians support close involvement in the performance of Da'wa in Muslim and non-Muslim countries. This is also one of the reasons why the politicians' stream is more popular among the younger generation, as they seek a more accessible discourse (Wiktorowicz Q (2006) Anatomy of the Salafi movement. Studies in Conflict & Terrorism, 29(3): 207–239.

2 In four schools of Sunni Islam (madhhab) Da'wa is associated with a soul searching (hisba) and with a verse from the Quran which states "commanding of good and forbidding of evil" (al-amr bi-l-maʿruf wa-n-nahy ʿan al-munkar). This is an ayah repeated several times in the Quran (3: 104, 110, 113,114).

3 For further information and example, refer to: Azzam A (1979) Defense of the Muslim Lands: The First Obligation After Iman. Islamic Books; Wagemakers, J. (2012). A quietist Jihadi: the ideology and influence of Abu Muhammad al-Maqdisi. Cambridge University Press; Ben-laden ("The ruling to kill the Americans and their allies—civilian and military—is an individual duty for every Muslim who can do it in any country in which it is possible to do it." In: "Declaration of the World Islamic Front for Jihad Against the Jews and Crusaders," 23 February 1998.

4 Rashid Rida, one of the founders of the Muslim Brotherhood and the editor-in-chief of the Egyptian magazine "Al-Manar" stated that Da'wa is obligatory in the magazine. In doing so, he inspired two important preachers; Hassan al-Banna, the organization's founder, and Muhammad Hamid al-Faqi, who founded the organization "Dar al-Da'wa and al-Arshad". Al-Bana gave Rida's ideas an organized and modern form for recruiting people. For him, the concept of Da'wa was a pillar in all aspects.

5 often referred to as the surface web.

6 https://www.cybersixgill.com/

7 الشيخ العلوان حفظه الله، تمنع الدين، آل سلول، لتكون كلمة الله هي العليا، صنم الوطنية، الديمقراطية صنم العصر، الولاء والبراء، ربانية حكيمة..

8 https://www.alarabiya.net/

9 BERT classification model: we used the pre-trained Arabic BERT model, called AraBERT (Safaya et al., 2020) and fine-tuned it to the classification task of detecting suspects. AraBERT is one of two highly used language models for the Arabic language (Williams et al., 2020). It was trained on the Arabic version of OSCAR (Open Super-large Crawled ALMAnaCH coRpus), on a recent dump of Arabic Wikipedia, and a number of other Arabic resources, a total of approximately 95GB of text. Given the diversity and the sources of this corpus, AraBERT's vocabulary contains Modern Standard Arabic, as well as some dialectical Arabic too.

10 This grammatical rule usually observed in literary Arabic.

11 An example of an MJD post of which at the end the user bases his words on a verse from the Quran: From the "al-Fidaa" forum on 06/06/2017.

الله درك و على الله أجرك اللهم منزل الكتاب ومجري السحاب وهازم الأحزاب، اهزم الروافض الحاقدين والصليبيين المتصهينين ومن حالفهم ..اللهم اجعلهم وعتادهم غنيمة للمسلمين.. اللهم دمّرهم وزلزلهم. اللهم أنت عضدنا وأنت نصيرنا ، اللهم بك نصول وبك نجول ..وبك نقاتل.. والله أكبر {وَلِلَّهِ الْعِزَّةُ وَلِرَسُولِهِ وَلِلْمُؤْمِنِينَ وَلَكِنَّ الْمُنَافِقِينَ لَا يَعْلَمُونَ} لا تنسونا من الدعاء.

May Allah bless you, and may your reward be with Allah. O Allah, Revealer of the Book, Mover of the clouds, and Defeater of the confederates, defeat the spiteful Rawafid (a derogatory term used for Shia Muslims), the Zionist Crusaders, and their allies… O Allah, make them and their equipment spoils for the Muslims. O Allah, destroy them and shake them. O Allah, You are our support, and You are our helper. O Allah, with You we strike, with You we move, and with You we fight… Allah is the Greatest. "And to Allah belongs [all] honor, and to His Messenger, and to the believers, but the hypocrites do not know. Do not forget us in your prayers" (Sura 63:8).

12 Scraped from http://www.al-eman.com.

13 A keyword given to a computer science in order to identify militias:

نزاع ,غزْوة , مليشيا, مليشيا البيشمركة, مليشيا الحشد, فرسان الثغور, مؤسسة البتار , ولاية الفرات, ولاية صلاح الدين, ولاية دجلة, ولاية الخير, ولاية خراسان, الرقة.

14 An example of a "real time" map of the occurrence of attack: https://twitter.com/ragipsoylu/status/1181091389737705473/photo/1-

15 An example of a post where the user indicates a large number of places in Syria: Forum Leyoth" 08/02/2015:

مقتل العشرات من الجنود الروافض قنصا وتفجيرا وتدمير عدد كبير من الآليات والثكنات في الموصل والفلوجة والحضر وديالى والأنبار. الله اكبر ولله الحمد.

The killing of dozens of Rawafid (a derogatory term for Shia Muslims) soldiers through sniper fire and explosions, and the destruction of a large number of vehicles and military barracks in Mosul, Fallujah, Hader, Diyala, and Anbar. Allah is the Greatest, and to Allah belongs all praise.

16 Words that characterize news items: ال : المصدر , عاجل, وكلة , ذكر ان, بيان, نقل.. Also, punctuation indicating news can be an overuse of colons, or ||.

17 The model was trained on "Amaq News Agency", a famous news agency of ISIS that was closed compared to Al-Arabiya newspaper, https://www.alarabiya.net, a recognized news agency based in Dubai.

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Author contributions
The idea for the study was conceived by Ms. Dafna Lewinsky. Dr. Yossi Mann assisted the team in refining the research question. The methodological framework was developed by Prof. Dov Te'eni and Prof. David G. Schwartz. Prof. Inbal Yahav addressed the technical aspects, helping the team translate human insights into machine outputs. Dr. Gahl Silverman supported the domain experts in creating a detailed methodology and provided feedback on earlier versions of the manuscript. All authors read and approved the final manuscript. All authors contributed to the manuscript revision and approved the submitted version.

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Competing interests
On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethical approval
Ethical approval was not sought for this study as data is derived from publicly available sources. Furthermore, the data was collected via an API from a third-party source, Sixgill, a company that provides CTI services and regularly monitors the Darknet for its clients. Additionally, no information in the data could expose people's personal details. Data that potentially could be used to assume the writer's identity based on a pseudonym was eliminated from the analyses and does not appear in the dataset. The focus of this study is ideological/thematic content of the posts.

Informed consent
Informed consent was not sought for this study since public data was collected via an API from a third-party source (Sixgill). The authors do not possess or have been exposed to any personal information.

Additional information
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